

STEM Faculty Perspectives: Assessment Design in the Era of Generative AI

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Abstract— This full research study explored STEM faculty perspective on the challenges they face in designing assessments and the strategies they used to mitigate its misuse in this new era of Generative artificial intelligence (GenAI). Specifically, the study aims to answer this research question: How do faculty members navigate the challenges of designing assessments in the context of GenAI tools? A qualitative approach was employed to answer our research questions. Semi-structured interviews were conducted with five STEM faculty focusing on their challenges, their approaches, and ethical concerns to safeguarding students against the unethical use of the GenAI tools while designing the classroom assessments. Thematic analysis revealed six major themes: 1) reimagining assessment in the age of GenAI, 2) perceived pedagogical potential of GenAI, 3) navigating ethical uncertainty in AI usage, 4) ambivalence and uneven adoption across faculty, 5) AI and educational equity, 6) envisioning a future-ready assessment culture. This study's findings contributed to the current discussion on the faculty's concerns on the students' unethical use of the GenAI tool while designing assessments. It is expected that this study, in combination with the subsequent studies, could be used to strengthen policies for ethical AI adoption in higher education assessment.

Keywords—GenAI, STEM, Ethics, Plagiarism

I. INTRODUCTION

Generative Artificial Intelligence (GenAI) tools have significantly altered the education landscape, especially the foundational assumption of learning, teaching, and assessment in the classroom [1]. In STEM education, educators, on the one hand, like the opportunities that the GenAI tools have created for them such as reduced workload, as they were able to create instructional material, create many problems, and develop classes plans quickly [2]. Also, GenAI are being used to inform their pedagogical practices [3]. On the other end, they are facing challenges in developing assessments to authentically evaluate the students learning and inform their instructional practice in the classroom [4]. The reason is that it is hard for instructors to differentiate between the work done by the students or by the GenAI tools in the absence of any standard plagiarism tool [5]. Also, traditional assessments such as multiple-choice questions (MCQs), open-ended questions, take-home exams, and problem-solving exercises have become irrelevant as students can easily generate solutions using GenAI tools [6]. Therefore, researchers are looking for ways to find alternative assessment strategies that are resilient to the use of GenAI.

This needs to rethink and redesign assessment to efficiently evaluate students' learning effectively has put a lot of pressure on the faculty. This concern is particularly urgent in STEM fields, where AI can reproduce technically correct but contextually shallow solutions [7], resulting in a lack of conceptual mastery by students. As a result, the core function

of assessment to inform instruction and promote learning is at risk. Therefore, there is a need to explore alternative approaches that could help instructors adopt these new challenges in developing the classroom assessment.

Recent literature has explored the impact of the GenAI tools on education, especially in terms of assessment. The focus of these studies is to understand the concerns related to academic integrity [8], [9], and the limitations of AI detection tools [5], [10]. However, there is a limited discussion on the assessment approaches used to mitigate or discourage unethical use of GenAI tool. One effective approach used in the literature is to evaluate a student's ability to apply knowledge in context [11], rather than simply recall or generate information. These include oral exams [12], live presentations [13], or project-based assessments [14] where the process, context, or collaboration are as important as the final product. Also, other resilient strategies used discussed in literature are those where students critique or improve AI-generated content, thereby demonstrating higher-order thinking [8]. These evolving assessment practices reflect a growing need not just to detect GenAI use but to design learning experiences that integrate it productively and ethically.

To contribute to the current efforts to fill this gap, this study aims to explore how STEM faculty are adapting their assessment design practices in light of the emergence of generative AI tools. Specifically, the study investigates this research question:

How do faculty members navigate the challenges of designing assessments in the context of Generative AI tools?

- How have faculty changed their assessments since the introduction of tools like ChatGPT?
- What types of assessment strategies are perceived to be more effective or resilient in the GenAI era?

To address these questions, this study employed a qualitative methodology to understanding STEM faculty experiences while desining their classroom assesement. This research aims to inform institutional policy, instructional design, and faculty development initiatives.

II. LITERATURE REVIEW

The emergence of GenAI tools such as ChatGPT has resulted in a significant shift in education by revolutionizing teaching methods, learning experiences, and institutional practices. These GenAI tools have the ability to produce human-like responses such as text, videos, and images to any query [15]. As a result, these tools are increasingly used by both students and instructors. Students have used these tools in many ways, such as brainstorming ideas [16], code

assistance [17], and getting feedback on assignments [18], etc. At the same time, the teachers have used these tools to generate instructional materials, engage in professional development, automatically grading the students assignments, and provide instant feedback to the students [2]. Despite these benefits, educators have raised serious concerns about the validity of assessment and academic integrity with the increasing use of AI by students in the assessments [9]. A key challenge identified is plagiarism in using GenAI tools for assessment in higher education [19].

A. Shifting Pedagogical Mindsets in Assessment Design:

Assessment plays a vital role in education by providing insights into student learning and guiding instructional strategies [20]. Assessment can be defined as any graded classroom activity that measures students' mastery of knowledge and/or skills [21]. Traditionally, instruction in the engineering classroom has primarily used different assessments, such as closed-ended problems, standardized exams, code evaluations, etc., to assess the student's understanding and learning of the taught concepts [22]. In these conventional assessment strategies, teachers act as evaluators and focus primarily on the final product rather than the process, emphasizing the correctness, efficiency, and objectivity of the final solution. However, this way of assessing student learning has become difficult in the age of Generative AI tools such as ChatGPT as these tools can produce human-like solutions (e.g., plausible code and essay). Hence, the researcher and educator are questioning the reliability of current assessment strategies [4], [9].

Current assessment strategies struggle to evaluate students' cognitive effort, originality, and authorship due to the prevalence of GenAI tools in the learning ecosystem. For instance, [7] conducted a study where they examined students' perceptions and use of GenAI powered chatbots for assessments. They found that over a third of students used AI chatbots for assistance with assessments, often not perceiving this as a breach of academic integrity. Therefore, educators and researchers are beginning to acknowledge that the traditional assessment models may not work where students retrieve and reproduce learned information [23]. As a result, instructors have started to adopt process-oriented assessments, where they assess the process used by the students to arrive at that solution [24]. Some examples of such assessments are annotated design processes, iterative design tasks, think-aloud protocols, etc. Overall, the mindset behind assessment is changing, and there has been a huge debate about finding a way to assess authentic student learning.

B. Assessment Challenges Faced by Educators in the GenAI Era

With the ever-evolving GenAI tools, the main challenge for the instructor in the assessments is to ensure the academic integrity of students' work. The educator has raised concerns that there are no reliable plagiarism tools to verify if the students' submitted work during assessment is their own or done by some GenAI tools such as ChatGPT [5]. This ambiguity places educators in a difficult position as there is no reliable way to verify the originality of student submissions. For instance, Myers [10] discussed the unreliability of AI detectors, especially when it comes to non-

native English workers whose legitimate work is often flagged as AI-generated. Another challenge is that most institutions lack clear and coherent policies and have a cautious, wait-and-see approach to AI adoption [25]. Hence, the instructors are often left to case-by-case decisions about acceptable AI use without such institutional policies. This causes inconsistency at various levels such as course, department, and college. Some instructors feel it is unfair to grade any work when some students disclose their use and others do not [26]. Others fear that over-reliance on AI tools could impede students' development of fundamental engineering skills [17].

Furthermore, studies have also pointed out that educator needs continuous professional development to stay updated with the evolving landscape of GenAI development [27], especially for the assessment. Redesigning assessments to be AI-resilient requires significant time and effort, especially in large or resource-constrained courses. The lack of institutional infrastructure for supporting these transitions further compounds the issue, leaving many educators overwhelmed and underprepared [27].

C. Strategies Adopted to Address Challenges

To address these challenges, instructors are exploring alternative assessment strategies that can be used to discourage the unethical use of the GenAI tools. Some alternative assessment strategies include the use of process-oriented (i.e., assess the process rather than the final solution to the assessment), such as oral exams (e.g., [12]), and live presentation (e.g., [13]), etc. In all these assessments, the core idea is to evaluate students' understanding directly, limiting the potential for unmonitored AI involvement. Another approach used to safeguard the unethical use of AI in assessment is to integrate the AI literature into the curriculum itself. This way, educator can foster ethical and responsible use of AI by equipping the with a comprehensive understanding of AI technologies, their capabilities, and ethical considerations. For instance, Hill and Hargis [28] integrated an ethics module on academic integrity and generative AI. By integrating this module, they involved students in the creation of AI usage policies and fostering dialogue around appropriate use, transparency, and assessment standards. This collaborative model not only gives students a sense of ownership but also highlights the value of integrity and personal responsibility in their professional development.

Another approach used in the literature to address the unethical use of AI in assessment is to increase its transparency in the evaluation. For instance, Furze and colleagues [8] discussed their AI Assessment Scale (AIAS) scale which allows educators to integrate AI into assessments while requiring students to transparently showcase their AI usage. They found that its implementation has resulted in reducing academic misconduct and improved student learning outcomes. Despite a growing body of commentary, empirical research on how STEM educators are responding to GenAI remains limited. Most current literature focuses on speculative risks or student behavior, leaving a gap in understanding the subjective experiences and practical responses of educators.

III. METHOD

A qualitative approach was employed to thoroughly understand the STEM faculty's perspective and their experience while designing the assessment for their class and the strategies they use to safeguard the unethical use of the GenAI.

A. Participants

For this Pilot study, we recruited 5 STEM faculty members from R1 (very high research activity) and R2 (high research activity) public universities across the different regions of the United States, bringing a diverse perspective. We used a covenant sampling approach to require these participants as they were easily accessible and met the study's inclusion criteria. The inclusion criteria were that the faculty should be teaching a STEM course and have some awareness of the GenAI tool in an educational setting. This approach allows us to recruit participants easily while remaining aligned with the study's focus. Faculty demographics are shown in Table 1.

TABLE I. DEMOGRAPHIC DISTRIBUTION OF THE PARTICIPANTS

Aliases	Gender identity	Teaching Area	Years of experience	University type
Ethan	Male	Computer science	~2	R1
Jinwoo	Male	Computer Science	~5	R1
Omar	Male	Engineering & Computer Science	15	R1
Imran	Male	Computer Science	>5	R1
Zayan	Male	Civil Engineering	~4	R2

B. Data Collection

We conducted Semi-structured interviews via Zoom, each lasting approximately 45 minutes. The interview was focused on collecting instructors' experiences with GenAI, their concerns about its unethical use in assessment, and the strategies they employ to mitigate AI misuse. Faculty in each interview were informed and asked for consent before recording the Zoom session. After the session, we transcribed the session using AI-powered, freely available transcription software. The AI-generated transcripts were verified by the researchers by randomly comparing the part of the video with the textual transcription.

C. Data Analysis

We analyzed the faculty interview data using thematic analysis following Braun and Clarke's [29] six-phase approach. We use an inductive coding approach to allow themes to emerge via interpretation based on the interview data. For that, we read the interview's transcripts multiple times to become familiar with the content of the content after transcription. After getting familiar with the data, we generated initial codes that reflect importantly align with our research questions by assigning them to phrases or sentences in the transcript. Then, these initial codes are categorized into broader themes reflecting participants' concerns and strategies related to the use of generative AI in assessments. Also, themes were reviewed and re-defined to ensure that they accurately captured the meanings of the data. In the end, we develop a narrative to explain each theme with supporting quotations, offering insights into faculty concerns and

mitigation strategies regarding GenAI in STEM assessment. To enhance the reliability of the analysis, two researchers independently coded a subset of the transcripts and compared results to establish inter-coder consistency.

IV. RESULTS

A. Theme 1: Reimagining Assessment in the Age of GenAI

Faculty reported actively reshaping assessment practices in response to the challenges and possibilities introduced by Generative AI. Moving away from traditional, easily compromised formats, they are increasingly adopting pedagogical strategies such as problem-based learning, flipped classrooms, oral interviews, and reflective assignments. These approaches are aimed to foster deeper engagement, minimize academic dishonesty, and ensure that the assessments capture students' genuine understanding.

One faculty member described their use of informal formative assessments to encourage engagement and discourage reliance on AI: "I begin class with informal formative assessment conversations, asking students about the previous topic or assigned video. Sometimes I pick teams randomly to answer, which encourages collaborative discussion and motivates students to review the material beforehand."

Another instructor discussed using GenAI to enhance teaching efficiency: "Primarily, I use AI for lecture preparation and sometimes to develop evaluation scenarios or test questions."

These practices illustrate a shift toward dynamic, interactive, and student-centered assessment approaches that not only deter unethical AI use but also promote active learning and concept retention.

B. Perceived Pedagogical Potential of GenAI

While faculty expressed caution around academic integrity, many also view Generative AI as a tool with considerable pedagogical value. Participants perceived GenAI not as a replacement for human instruction, but as a complementary resource that, when used thoughtfully, could enhance student learning, and promote efficiency.

One participant emphasized that GenAI benefits students who already possess foundational knowledge:

"These tools are really useful for people who already have the base knowledge. But for students starting from scratch, there's a risk they may over-rely on AI."

Another faculty member described how they frame GenAI as a productivity tool that encourages active inquiry:

"I tell students: if you're confused about a concept, use AI to explain it in simpler terms like asking it to teach as if to a high school student. I even share prompts with them to help guide their learning."

These reflections suggest that faculty see GenAI as a scaffold, that when used appropriately, can help students clarify misunderstandings, supplement instruction, and build academic confidence.

C. Navigating Ethical Uncertainty in AI Usage

As GenAI tools become increasingly accessible, faculty face the complex challenge of maintaining academic integrity while accommodating new technologies. Among the faculty participants, while some have started introducing ethical

guidelines in their classrooms, many remain uncertain about where the boundaries of appropriate AI use should be drawn.

One instructor shared their flexible but reflective approach:

“At the start of class, I tell students they can use these tools, but I’m interested in seeing whether they actually understand what they’ve done with them.”

This permissive stance focused on students’ understanding of the concepts instead of outright prohibition, suggesting an evolving standard of accountability.

Yet, even as some faculty attempts to instill ethical practices, there is widespread ambiguity about what constitutes misuse:

“When we say something is unethical, especially after allowing students to use AI, drawing that line becomes really hard.”

These uncertainties show the tension that faculty face between supporting students’ learning and upholding the academic standards. Faculty are aware of the need to guide responsible use, but they often lack shared norms or institutional clarity to do so.

D. Ambivalence and Uneven Adoption Across Faculty

Faculty engagement with GenAI tools varied significantly, reflecting differences in technological familiarity, pedagogical priorities, and comfort with innovation. While some instructors experimented with AI in their assessment design, others remained skeptical or limited their use due to unfamiliarity. One participant explained:

“I would say it's sometimes. I build the use cases myself and then ask ChatGPT just to see how well I can test the student based on their knowledge. So, I will only get the questions out of it... not overly relying on it, but just wherever you need. I just need to get a buddy to look at it and see if I’m doing right or maybe create some new content.”

Another reflected more candidly on their limited use:

“Yep, ChatGPT. And since I’m kind of an old person, I’m not very much familiar with many tools, but sometimes I use ChatGPT myself to draft an email idea for me. So yeah, I know about it.”

These examples illustrate a spectrum of adoption from cautious use to minimal engagement indicating that faculty perceptions of GenAI are still forming and are often shaped by individual experiences and exposure.

E. AI and Educational Equity

Faculty raised important concerns about the unequal access to GenAI tools among students, particularly those from marginalized or international backgrounds. While some instructors saw access as less of a problem at their institutions, others viewed it as a pressing equity issue that institutions must address proactively. One participant reflected on the interplay between human and AI biases and how access is mediated by institutional support:

“We humans have our own biases. AI has its own. So, I mean that is one thing which we still have the advantage on as a human, you can guess that and come up with some strategy based on the case instead of something generic. It is hard at this stage, both in terms of seeing if they don't have access to [AI], because we at [university] provide them with laptops and internet access. So even if they don't have it at home, it would be hard...”

Another participant emphasized that institutional responsibility is key to ensuring fair use of GenAI:

“It's an equity issue. Somebody can pay, somebody cannot pay for the premium version of ChatGPT, right? ... Dealing with this situation is very tricky. But I think that is one of the reasons we should manage the use of AI tools... If we are allowing the use of it, we should enable everyone and provide support to everyone... We should make these tools available to everyone and then manage the use of it instead of being against it.”

He continued: “We should encourage it, and we should train both the educators, the assessment designers, and really students in a better use of these tools.”

This theme underscores the tension between encouraging innovation and maintaining fairness, particularly in environments where access to digital tools is uneven and support infrastructure is not uniformly in place.

F. Envisioning a Future-Ready Assessment Culture

Faculty also shared forward-thinking perspectives on how assessment practices must evolve not only to integrate GenAI, but to support students' ethical development, self-regulation, and conceptual understanding. Their reflections emphasized the need for cultural and pedagogical shifts that go beyond technological adjustments. One participant spoke about promoting healthy help-seeking behaviors:

“One of the things that I would like to mention here is we need to work on students’ healthy help-seeking behaviors... They won’t have us for 24 hours with them... So, the use of AI tools is not bad. This use has to be healthy. It has to be used in a healthy way. So, training students in those self-regulatory skills where they can learn these help-seeking behaviors can optimize the use of AI tools in their learning”

Another faculty emphasized the importance of prioritizing understanding over deadlines:

“As soon as I see they don't have the understanding of things, but they met the deadline, and they submitted it... I generate a rubric where most of the reward will come from the understanding part of it than the actual submission... And then another thing is, if they're not happy with the way they are rewarded, then I'll ask them, give it another try... but this time you should have the understanding of it.”

These comments suggest that some faculty have already shifted towards formative, flexible, and mastery-oriented approaches, where learning is emphasized over performance, and students are encouraged to engage in iterative improvement.

V. DISCUSSION

In this study, we explored the faculty's perspective on their concerns, strategies, and experience in designing assessments in this era of Generative AI. One major theme from this study is that faculty are facing challenges while designing assessments that can effectively evaluate authentic student learning with the increasing use of generative AI (GenAI) tools. Therefore, faculty are reimagining assessment practices resistant to GenAI tool usage. In this regard, the faculty believes that traditional assessments, such as take-home essays and standardized tests, are increasingly vulnerable to AI-generated responses. This finding aligns with the current discussion happening on the re-evaluation of our assessment [8], [9], [23]. In response to the shortcomings

of previous or traditional assessments, faculty are exploring alternate assessment strategies that require high-order things or have real-time demonstrations (e.g., oral exams) of learning. This shift in the assessment shows a broader discussion suggesting the need for adaptable assessments and pedagogies to adapt to this new trend of AI rather than resisting it. These findings also align with previous studies, suggesting that in this GenAI era, assessing high-order thinking or in-class demonstration of learning might be a better approach to understanding the authentic student mastery of knowledge or skills and informing their instructional practices [30], [31] [12].

While educators are concerned about GenAI misuse, another theme emerges from the faculty interview: using GenAI in the pedagogical design has a potential benefit. For instance, faculty believed that the GenAI has the potential to act as a personalized tutor to provide immediate scaffolding, help in brainstorming, and provide learning support. Faculty who saw value in GenAI emphasized that it can promote student engagement and foster deeper learning if use responsibility and ethically by students [32]. Another significant theme is the uneven adoption of AI among faculty members. While some faculty advocated the integration of AI into all aspects of teaching, i.e., designing instruction, allowing students to use GenAI with proper adoption, and considering GenAI in terms of collaborative learning with peers, etc. Others also voice sliced skepticism and resistance, often due to lack of training, fear of not identifying the student's authentic work, or concerns over the excessive student dependence on GenAI usage while working on the assessment. This uneven adoption might be due to inconsistency in terms of the dialogues to build confidence among educators by the college leadership, policymakers, and even researchers [33].

Faculty also discussed the equality and access issues of GenAI usage and expertise. They believe that all students might not have equal access to the GenAI tools and digital literacy to effectively engage with them. This divide among students further complicates the issue of assessment. For instance, it becomes difficult to fairly evaluate students with expertise in AI usage and those without such experience without reliably assessing the authenticity of students' work.

Finally, faculty have shown consensus on the inevitability of AI's role in education and emphasized the importance of preparing faculty and students for its adoption. We need support for teachers to have ongoing professional development to stay up to date with AI development and need clear AI usage policies at the institutional level. For students, we must create opportunities as faculty to practice GenAI tools and become experts to prepare them for the industry. Overall, faculty members are actively engaged in discussions with their peers and literature to negotiate the challenges and possibilities provided by GenAI.

VI. CONCLUSION AND FUTURE DIRECTIONS

This study explored faculty perspectives on the concerns and mitigation strategies they use to safeguard against generative AI (GenAI) misuse in student assessments. The key findings revealed that most faculty members believe there is a need to re-evaluate our assessment strategies as the current assessment strategies are not resilient enough to prevent or discourage unethical AI use. In that regard, they

discuss several strategies they have employed to ensure students document the extent of their GenAI use while working on assessments. Furthermore, they discussed the need for continuous professional development and clear AI usage policies at the institutional level.

However, faculty still express concerns about the potential for academic dishonesty using GenAI tools, as it is challenging to distinguish AI-generated work from authentic student work. In response, they have shifted to assessments where it requires high-order things or have aspects of real-time demonstrations (e.g., oral exams) of learning. This approach is the only one they feel confident about in ensuring that students are not using AI, and they believe these assessments reflect authentic learning. The study also highlighted a significant need for institutional support, with faculty calling for clearer guidelines and professional development to navigate these challenges better.

The result of this study suggests that while the GenAI tool has helped instructors and students in many ways, its issues are still a huge concern among educators. This finding highlights the tension between technological advancement and academic integrity in higher education. At the same time, it also encourages us to re-evaluate our assessment. Furthermore, all instructors thought that the ethical use of technological advancement should be taught to the students to prepare them for the future workforce. However, it needs to be reiterated in the classroom and through their learning journey that this should act as your peer or a helping hand rather than replace your critical thinking, creativity, and problem-solving skills. This study contributes to the current discussion of practical challenges faced by faculty and the strategies they employ to maintain academic standards in this GenAI era.

Our future direction builds on the current study's limitations while also extending it. We intend to increase the sample size and add more diverse participants to make the study more generalizable. Another direction is to include the students' perspective, exploring students' motivations for using AI tools in assessments and their understanding of academic integrity. Also, as this was a cross-sectional study, we can design longitudinal studies to track the long-term effectiveness of the strategies employed by faculty to mitigate unethical GenAI use and assess their impact on student learning outcomes. Finally, we could also examine the effectiveness of AI-based detection tools, as they are becoming more advanced, and their role in maintaining academic standards.

REFERENCES

- [1] F. Han and Z. Wang, "Chinese university EFL learners' essay revision and perceptions of the feedback A comparison between teacher vs. ChatGPT feedback," *Asia-Pac. J. Educ. Arts Sci.*, 2024, doi: 10.14742/apubs.2024.1230.
- [2] E. Kasneci *et al.*, "ChatGPT for good? On opportunities and challenges of large language models for education," *Learn. Individ. Differ.*, vol. 103, p. 102274, Apr. 2023, doi: 10.1016/j.lindif.2023.102274.
- [3] A. A. Butt, E. Babar, M. Menekse, and A. Alhaddad, "A Comparative Analysis of Large Language Models and NLP Algorithms to Enhance Student Reflection Summaries," presented at the ASEE Annual Conference & Exposition, Portland, OR, 2024.
- [4] S. MacNeil, S. Spurlock, and I. Applebaum, "Imagining Computing Education Assessment after Generative AI," Jan. 2024, doi: 10.48550/arXiv.2401.04601.

- [5] S. Pudasaini, L. Miralles-Pechuán, D. Lillis, and M. Llorens Salvador, "Survey on Plagiarism Detection in Large Language Models: The Impact of ChatGPT and Gemini on Academic Integrity," Jun. 2024, doi: 10.48550/arXiv.2407.13105.
- [6] A. Chirumamilla, G. Sindre, and A. Nguyen-Duc, "Cheating in e-exams and paper exams: the perceptions of engineering students and teachers in Norway," *Assess. Eval. High. Educ.*, vol. 45, no. 7, pp. 940–957, Oct. 2020, doi: 10.1080/02602938.2020.1719975.
- [7] J. H. Gruenhagen, P. M. Sinclair, J.-A. Carroll, P. R. A. Baker, A. Wilson, and D. Demant, "The rapid rise of generative AI and its implications for academic integrity: Students' perceptions and use of chatbots for assistance with assessments," *Comput. Educ. Artif. Intell.*, vol. 7, p. 100273, Dec. 2024, doi: 10.1016/j.caeai.2024.100273.
- [8] L. Furze, M. Perkins, J. Roe, and J. MacVaugh, "The AI Assessment Scale (AIAS) in action: A pilot implementation of GenAI-supported assessment," *Australas. J. Educ. Technol.*, vol. 40, no. 4, pp. 38–55, Oct. 2024, doi: 10.14742/ajet.9434.
- [9] Z. Swiecki *et al.*, "Assessment in the age of artificial intelligence," *Comput. Educ. Artif. Intell.*, vol. 3, p. 100075, Jan. 2022, doi: 10.1016/j.caeai.2022.100075.
- [10] A. Myers, "AI-detectors biased against non-native English writers," *Stanf. HAI Retrieved March*, vol. 1, p. 2024, 2023.
- [11] F. J. Agbo, H. K. Smalley, and K. Nyman, "Competency-Based Assessment in the Era of Generative Artificial Intelligence: Perspectives of Selected STEM Educators," presented at the International Conference on Computers in Education, 2024. doi: 10.58459/icce.2024.4970.
- [12] G. J. Mariano, D. E. Allwardt, P. R. Raptis, and K. Stilwell, "Reintroducing the oral exam: Finding out what your students really know in the age of ChatGPT," *Curr. Teach. Learn.*, vol. 16, no. 1, pp. 59–69, Sep. 2024.
- [13] R. Mulder, C. Baik, and T. Ryan, "Rethinking Assessment in Response to AI," Melbourne Centre for the Study of Higher Education, Melbourne, Australia, Jul. 2023. [Online]. Available: https://melbourne-cshe.unimelb.edu.au/_data/assets/pdf_file/0004/4712062/Assessment-Guide-Web-Final.pdf
- [14] N. Boughattas, W. Neji, and F. Ziadi, "Project-Based Assessment in the Era of Generative AI: Challenges and Opportunities," presented at the Proceedings of the 20th International CDIO Conference, Tunis, Tunisia: ESPRIT School of Engineering, Jun. 2024, pp. 347–356.
- [15] F. Fui-Hoon Nah, Zheng ,Ruilin, Cai ,Jingyuan, Siau ,Keng, and L. and Chen, "Generative AI and ChatGPT: Applications, challenges, and AI-human collaboration," *J. Inf. Technol. Case Appl. Res.*, vol. 25, no. 3, pp. 277–304, Jul. 2023, doi: 10.1080/15228053.2023.2233814.
- [16] H. K. Kim *et al.*, "ChatGPT and me: Collaborative creativity in a group brainstorming with generative AI," presented at the 2024 ASEE Annual Conference & Exposition, Jun. 2024. [Online]. Available: <https://peer.asee.org/chatgpt-and-me-collaborative-creativity-in-a-group-brainstorming-with-generative-ai>
- [17] A. Ghimire and J. Edwards, "Coding with AI: How Are Tools Like ChatGPT Being Used by Students in Foundational Programming Courses," in *Artificial Intelligence in Education*, A. M. Olney, I.-A. Chounta, Z. Liu, O. C. Santos, and I. I. Bittencourt, Eds., Cham: Springer Nature Switzerland, 2024, pp. 259–267.
- [18] D. Sun, Cheng ,Gary, Yu ,Philip Leung Ho, Jia ,Jiyu, Zheng ,Zhizi, and A. and Chen, "Personalized stem education empowered by artificial intelligence: a comprehensive review and content analysis," *Interact. Learn. Environ.*, pp. 1–23, doi: 10.1080/10494820.2025.2462156.
- [19] D. R. E. Cotton, Cotton ,Peter A., and J. R. and Shipway, "Chatting and cheating: Ensuring academic integrity in the era of ChatGPT," *Innov. Educ. Teach. Int.*, vol. 61, no. 2, pp. 228–239, Mar. 2024, doi: 10.1080/14703297.2023.2190148.
- [20] X. Gao, P. Li, J. Shen, and H. Sun, "Reviewing assessment of student learning in interdisciplinary STEM education," *Int. J. STEM Educ.*, vol. 7, no. 1, p. 24, Jun. 2020, doi: 10.1186/s40594-020-00225-4.
- [21] E. Huber *et al.*, "Towards a framework for designing and evaluating online assessments in business education," *Assess. Eval. High. Educ.*, vol. 49, no. 1, pp. 102–116, Jan. 2024, doi: 10.1080/02602938.2023.2183487.
- [22] K. J. Chew and H. M. Matusovich, "Exploring fundamental engineering course instructors' test usage beliefs and behaviors: A multicase study," *J. Eng. Educ.*, vol. 113, no. 4, pp. 1146–1176, Oct. 2024, doi: 10.1002/jee.20614.
- [23] R. Kadel, B. K. Mishra, S. Shailendra, S. Abid, M. Rani, and S. P. Mahato, "Crafting Tomorrow's Evaluations: Assessment Design Strategies in the Era of Generative AI," May 2024, doi: 10.48550/arXiv.2405.01805.
- [24] T. H. Feng, A. Luxton-Reilly, B. C. Wünsche, and P. Denny, "From Automation to Cognition: Redefining the Roles of Educators and Generative AI in Computing Education," in *Proceedings of the 27th Australasian Computing Education Conference*, in ACE '25. New York, NY, USA: Association for Computing Machinery, 2025, pp. 164–171. doi: 10.1145/3716640.3716658.
- [25] Y. Jin, L. Yan, V. Echeverria, D. Gašević, and R. Martinez-Maldonado, "Generative AI in higher education: A global perspective of institutional adoption policies and guidelines," *Comput. Educ. Artif. Intell.*, vol. 8, p. 100348, Jun. 2025, doi: 10.1016/j.caeai.2024.100348.
- [26] J. Lundin, M. Utterberg Modén, T. Leino Lindell, and G. Fischer, "A Remedy to the Unfair Use of AI in Educational Settings," *Interact. Des. Archit. J. - IxDA*, vol. 59, no. Winter 2023-2024, pp. 62–78, Jan. 2024, doi: 10.55612/s-5002-059-002.
- [27] X. Tan, G. Cheng, and M. H. Ling, "Artificial intelligence in teaching and teacher professional development: A systematic review," *Comput. Educ. Artif. Intell.*, vol. 8, p. 100355, Jun. 2025, doi: 10.1016/j.caeai.2024.100355.
- [28] C. Hill and J. Hargis, "An ethics module on academic integrity and generative AI," *New Dir. Teach. Learn.*, vol. n/a, no. n/a, Sep. 2024, doi: 10.1002/tl.20626.
- [29] V. Braun and V. Clarke, "Using thematic analysis in psychology," *Qual. Res. Psychol.*, vol. 3, no. 2, pp. 77–101, Jan. 2006, doi: 10.1191/1478088706qp063oa.
- [30] P. Eachempati, R. Komattil, and A. Arakala, "Should oral examination be reimaged in the era of AI?," *Adv. Physiol. Educ.*, vol. 49, no. 1, pp. 208–209, 2025, doi: 10.1152/advan.00191.2024.
- [31] J. Pearce and N. Chiavaroli, "Prompting Candidates in Oral Assessment Contexts: A Taxonomy and Guiding Principles," *J. Med. Educ. Curric. Dev.*, 2020, doi: 10.1177/2382120520948881.
- [32] C. K. Y. Chan and K. K. W. Lee, "The AI generation gap: Are Gen Z students more interested in adopting generative AI such as ChatGPT in teaching and learning than their Gen X and millennial generation teachers?," *Smart Learn. Environ.*, vol. 10, no. 1, p. 60, Nov. 2023, doi: 10.1186/s40561-023-00269-3.
- [33] L. Hughes, T. Malik, S. Dettmer, A. S. Al-Busaidi, and Y. K. Dwivedi, "Reimagining Higher Education: Navigating the Challenges of Generative AI Adoption," *Inf. Syst. Front.*, Feb. 2025, doi: 10.1007/s10796-025-10582-6.